⇒ Rocket Traj. Sim. Exercise 8 Notes:

Extend the rocket traj. sim. from 1D to 2D, & apply an initial offset from vertical to initiate a gravity turn. Additionally, make plots of...

- Altitude vs downrange distance
- Flight path angle vs time
- Acceleration vs time
- Dynamic pressure vs time
- Mass vs time
- Trajectory & limb of Earth

> Initiate Gravity Turn

If a rocket is initially launched vertically, i.e., starts its motion along the direction of the position vector, then gravity turn must be initiated using a rotation. For example, a gravity turn trajectory in the x-y plane can be initiated using off-axial thrust modeled with a rotation about the z-axis:

$$\vec{a}_{thrust} = \frac{V_e \dot{m}}{M_{empty} + M_{propellant} - \dot{m}t} R_z(\theta)\,\hat{r}$$

where

$$\hat{r} = \frac{\vec{r}}{\|\vec{r}\|} = \frac{[r_x, r_y]}{\sqrt{r_x^2 + r_y^2}}, \quad R_z(\theta) = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}$$

Implementation in Python:

```
import numpy as np
R_z = np.array([np.cos(theta),-np.sin(theta), 0],
               [np.sin(theta), np.cos(theta), 0])

r = np.array([r_x], [r_y])
r_norm = np.linalg.norm(r)
a_thrust = T_magnitude * np.matmul(R_z,r)/r_norm
```

Ex. 8 Notes Cont.

> Rocket Equations of Motion

- Acceleration is given by:

$$\vec{a} = \vec{a}_{gravity} + \vec{a}_{thrust} + \vec{a}_{drag}$$

where

$$\vec{a}_{gravity} = -\frac{GM_{\oplus}}{||\vec{r}||^2}\hat{r}$$

$$\vec{a}_{thrust} = \frac{T}{m_{empty} + m_{propellant} - \dot{m}t}\hat{d}$$

$$\vec{a}_{drag} = -\frac{1}{2}\frac{\rho||\vec{v}_{rel}||^2 C_d A}{m_{empty} + m_{propellant} - \dot{m}t}\hat{d}$$

G = universal gravitational constant = $6.6743*10^{-11}$ [Nm^2/kg^2]

$M_{\oplus}$ = mass of Earth = $5.9722*10^{24}$ [kg]

$\vec{r}$ = position vector in inertial frame

$\vec{v}$ = velocity vector in inertial frame

$\vec{v}_{rel}$ = velocity relative to surface of Earth

$= \vec{v} - \vec{\omega}_{\oplus} \times \vec{r}$

$\vec{\omega}_{\oplus}$ = Earth's angular velocity vector

C_d = drag coefficient of rocket

A = cross sectional area of rocket

$\hat{d}$ = unit direction vector such that...

if $t < t_{pitch\ program\ start}$:

$\hat{d} = \hat{r} = \vec{r}/||\vec{r}||$ i.e. rocket moves along position vector

if $t \geq t_{pitch\ program\ start}$:

$\hat{d} = \hat{v} = \vec{v}/||\vec{v}||$ i.e. rocket moves along velocity vector

- Velocity is given by integrating acceleration once, i.e.,

$\vec{v} = \frac{d\vec{v}}{dt}dt = \vec{a}\,dt$

- Position is given by integrating velocity once, i.e.,

$\vec{r} = \frac{d\vec{r}}{dt}dt = \vec{v}\,dt$

Ex. 8 Notes Cont.

> Rocket 2D Eqns of Motion:

$$\vec{a} = a_x \hat{i} + a_y \hat{j}$$

where

$$a_x = -\frac{GM_\oplus r_x}{\sqrt{r_x^2 + r_y^2}^3}$$

$$\ldots + \frac{V_e \dot{m}}{M_{empty} + M_{propellant} - \dot{m}t} d_x$$

$$\ldots - \frac{1}{2} \frac{\rho \sqrt{V_{rel,x}^2 + V_{rel,y}^2}\, C_d A}{M_{empty} + M_{propellant} - \dot{m}t} d_x$$

$$a_y = -\frac{GM_\oplus r_y}{\sqrt{r_x^2 + r_y^2}^3}$$

$$\ldots + \frac{V_e \dot{m}}{M_{empty} + M_{propellant} - \dot{m}t} d_y$$

$$\ldots - \frac{1}{2} \frac{\rho \sqrt{V_{rel,x}^2 + V_{rel,y}^2}\, C_d A}{M_{empty} + M_{propellant} - \dot{m}t} d_y$$

where

d_x, d_y = direction coefficients such that...

if $t < t_{pitch\ program\ start}$:

$$d_x = r_x / \sqrt{r_x^2 + r_y^2}, \quad d_y = r_y / \sqrt{r_x^2 + r_y^2}$$

if $t \geq t_{pitch\ program\ start}$:

$$d_x = V_x / \sqrt{V_x^2 + V_y^2}, \quad d_y = V_y / \sqrt{V_x^2 + V_y^2}$$

To solve for position & velocity of rocket throughout its trajectory, must solve the nonlinear system of ODEs:

$$\frac{d}{dt}\begin{pmatrix} r_x \\ r_y \\ V_x \\ V_y \end{pmatrix} = \begin{pmatrix} V_x \\ V_y \\ a_x \\ a_y \end{pmatrix}$$

Solved via numerical integration (Euler, Midpoint, RK4, etc.)!!!

Ex. 8 Notes Cont.

> Definitions for Plots:

- Dynamic pressure:

  $$q = \tfrac{1}{2} \rho v_{rel}^2$$

  where

  ρ = air density [kg/m³]

  v_{rel} = velocity relative to surface of Earth

  $= \vec{v} - \vec{\omega}_{\oplus} \times \vec{r}$

  $\vec{\omega}_{\oplus}$ = Earth's angular velocity vector

- Flight Path Angle:

  $$\gamma = \frac{\pi}{2} - a\cos\left(\frac{\vec{r} \cdot \vec{v}}{\|\vec{r}\| \, \|\vec{v}\|}\right)$$

  $$= a\sin\left(\frac{\vec{r} \cdot \vec{v}}{\|\vec{r}\| \, \|\vec{v}\|}\right)$$

Exercise 8

Review of

⇒ Rotation Matrices:

- about x-axis:

$$C_x(a) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(a) & -\sin(a) \\ 0 & \sin(a) & \cos(a) \end{bmatrix}$$

} Memorize this, then others are just a cyclic permutation of this

- about y-axis:

$$C_y(a) = \begin{bmatrix} \cos(a) & 0 & \sin(a) \\ 0 & 1 & 0 \\ -\sin(a) & 0 & \cos(a) \end{bmatrix}$$

- about z-axis:

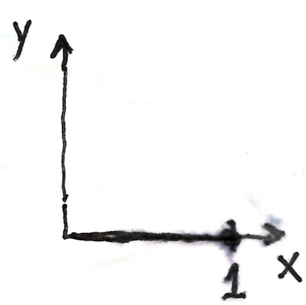
$$C_z(a) = \begin{bmatrix} \cos(a) & -\sin(a) & 0 \\ \sin(a) & \cos(a) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

- Example: rotate vector [1, 0, 0] about z-axis by 45°

$V_0 = [1, 0, 0]$, $a = \pi/4$

$$V_1 = C_z(\pi/4)\, V_0 = \begin{bmatrix} \cos(\pi/4) & -\sin(\pi/4) & 0 \\ \sin(\pi/4) & \cos(\pi/4) & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} \cos(\pi/4) \\ \sin(\pi/4) \\ 0 \end{bmatrix} = \begin{bmatrix} \sqrt{2}/2 \\ \sqrt{2}/2 \\ 0 \end{bmatrix}$$



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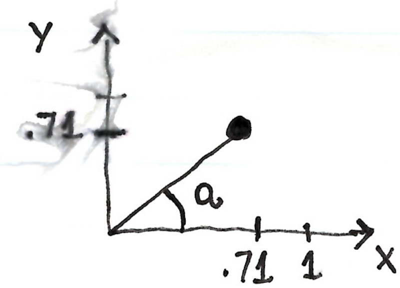
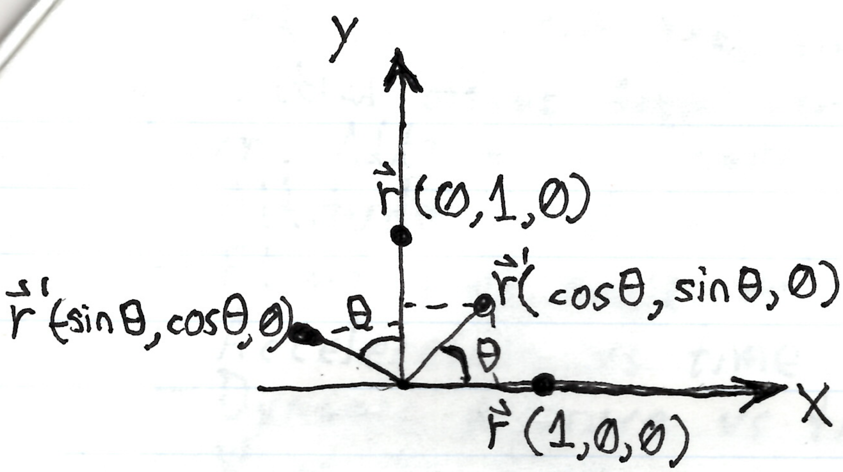


erivation of Rotation Matrix R_z:



$$\vec{r}' = R_z \vec{r}$$

where

$$R_z = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Good derivation at...
[robotacademy.net.au/lesson/describing-rotation-in-2d/]